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%% Classify-X: Object Classification using Mobile Camera
% Classifies 3 images using two pretrained neural networks:
% GoogLeNet and SqueezeNet
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Connect to MATLAB Mobile

```
clear m;
m = mobiledev;
clear cam;
cam = camera(m, 'back');
```

```
%% Load pretrained neural networks
[net1,classNames1] = imagePretrainedNetwork("googlenet");
[net2,classNames2] = imagePretrainedNetwork("squeezeNet");
```

```
%% Classify 3 images
for i = 1:3

    % Capture image from mobile camera
    fprintf("\n=====\\n");
    fprintf("Capture Image %d of 3\\n",i);
    fprintf("=====\\n");

    img = snapshot(cam, 'manual');

    % Display captured image
    figure;
    imshow(img);
    title("Captured Image " + i);

    %% Classify using GoogLeNet
    inputSize1 = net1.Layers(1).InputSize;

    img1 = imresize(img,inputSize1(1:2));
    img1 = single(img1);

    scores1 = predict(net1,img1);

    label1 = scores2label(scores1,classNames1);

    %% Classify using SqueezeNet
    inputSize2 = net2.Layers(1).InputSize;

    img2 = imresize(img,inputSize2(1:2));
    img2 = single(img2);

    scores2 = predict(net2,img2);
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label2 = scores2label(scores2,classNames2);

%% Display classification results
fprintf("Image %d Classification Results:\n",i);
fprintf("GoogLeNet  : %s\n",string(label1));
fprintf("SqueezeNet  : %s\n",string(label2));

title("GoogLeNet: " + string(label1) + ...
      " | SqueezeNet: " + string(label2));

end

```

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=====
Capture Image 1 of 3
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Image 1 Classification Results:
GoogLeNet  : water bottle
SqueezeNet  : water bottle

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GoogLeNet: water bottle | SqueezeNet: water bottle



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=====
Capture Image 2 of 3
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Image 2 Classification Results:
GoogLeNet  : screwdriver
SqueezeNet  : spindle

```

GoogLeNet: screwdriver | SqueezeNet: spindle



=====
Capture Image 3 of 3
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Image 3 Classification Results:

GoogLeNet : modem

SqueezeNet : mouse

GoogLeNet: modem | SqueezeNet: mouse

